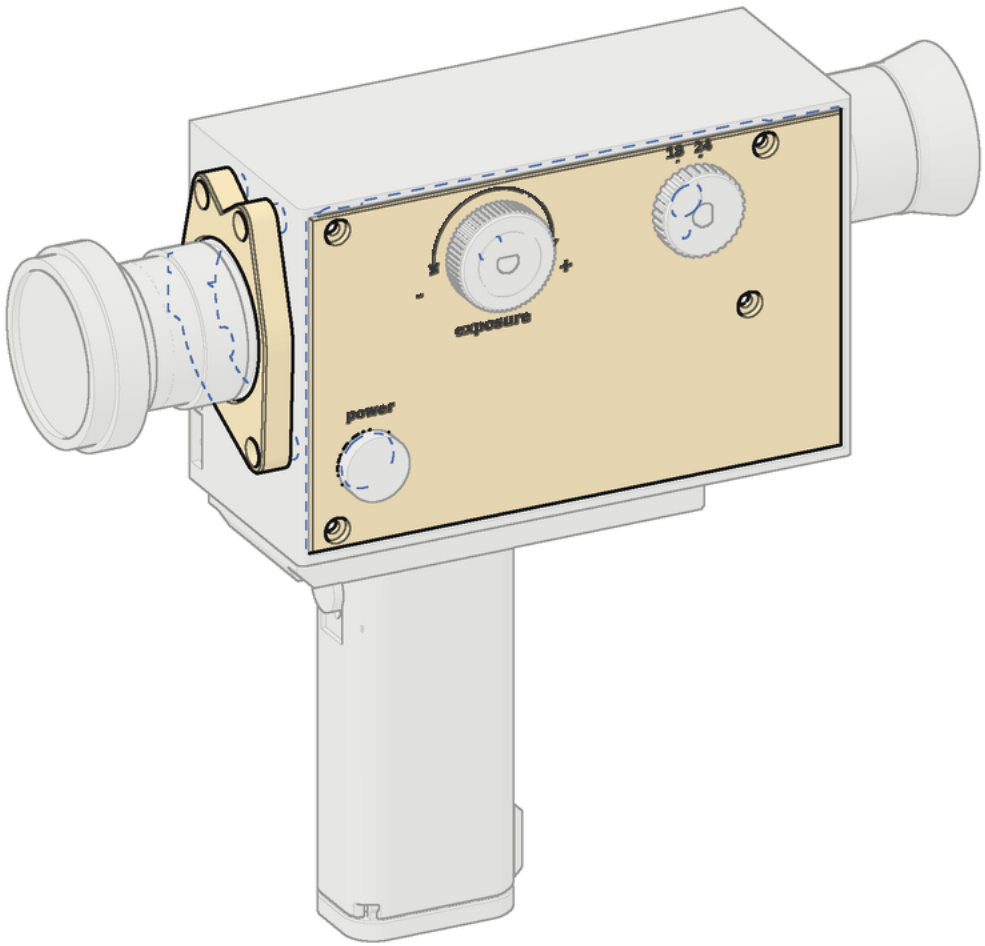
D2 2026-10-12 corrective assembly review

BLOCKED REVIEW | Vertical Pi candidate | 158 x 63.6 x 100 mm core | 10 physical prints / 9 CAD rows



A sequence to test

Pictures expose fit, insertion and service dependencies.

Amber: new part.
Grey: already present.
Blue: proposed motion.
Dashed: hidden part.

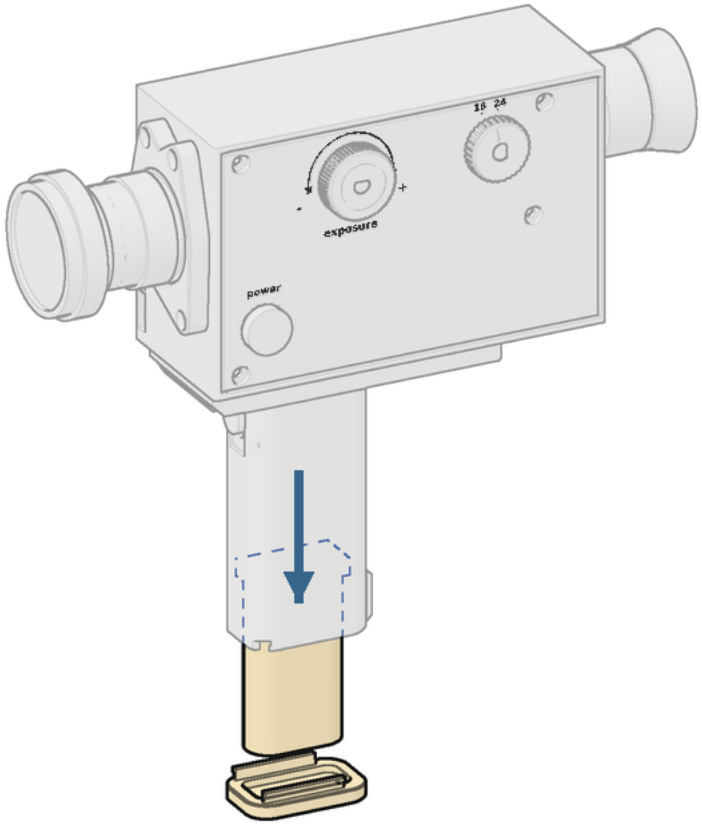
The outer plenum is integral to the tub.

No physical build is qualified.

STOP All 56 inherited physical gates and M1 remain open. Do not fabricate or energize from this review.

00 Disconnect before opening

The J2 button requests power control. It does not isolate the battery.



Four prerequisites

1 Stop recording; clean shutdown.

2 Remove every external supply.

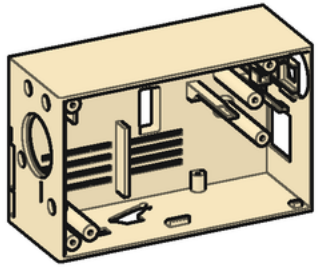
3 Withdraw pack; part XT30 by housings. Protect contacts.

4 Only then touch internal connectors. Use ESD protection. Charge the pack outside the body.

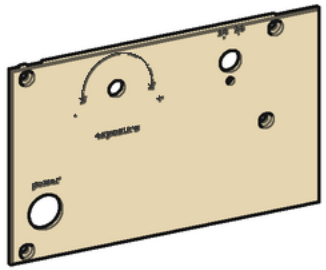
STOP XT30 must be reachable with body closed and retention fitted. Fixed tied loops cannot unfold for service.

01 Count the printed parts

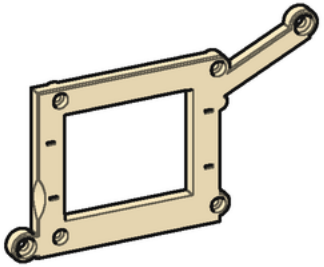
Compact: 10 physical prints in 9 CAD assembly rows. Body and bonded catch stub are separate prints. Gauge / coupons excluded.



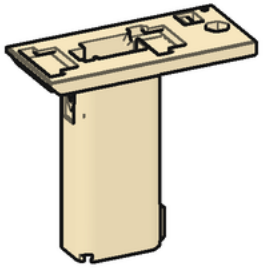
1 + 2 Body + bonded stub



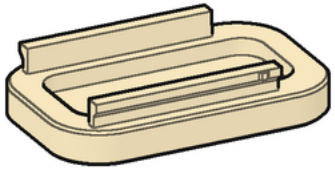
3 Panel



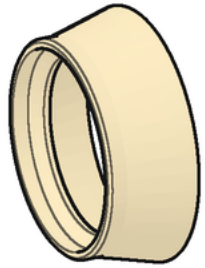
4 Carrier



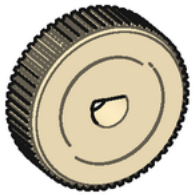
5 Base / grip



6 Battery cap



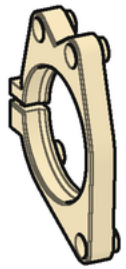
7 Eyecup



8 Exposure knob



9 18 / 24 knob

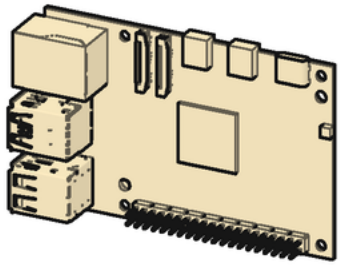


10 Lens collar

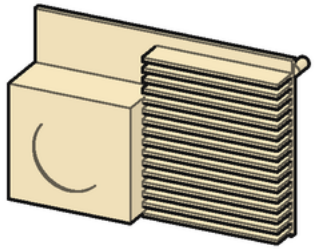
Tub shown assembled above; separate print pieces on page 03x. Original storage adds one sleeve: 11 physical prints / 10 CAD rows.

01b Identify the purchased assemblies

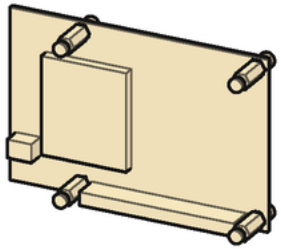
CAD envelopes. Match actual purchased identities and revisions; each group is one installed assembly.



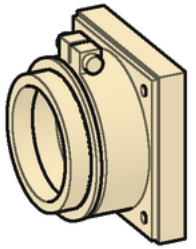
A Raspberry Pi 5



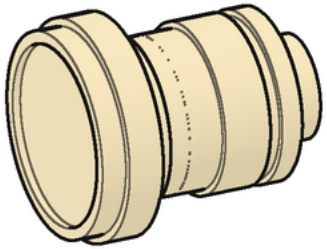
B Active Cooler envelope



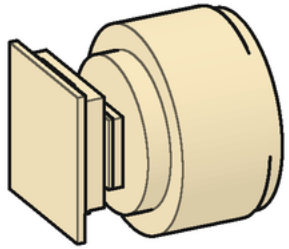
C X1203 + kit hardware



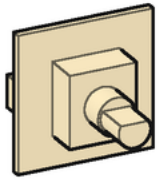
D GS camera + C-CS



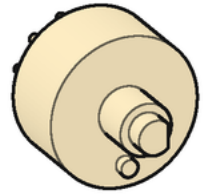
E Kowa LM6HC lens



F EVF kit (flex not shown)



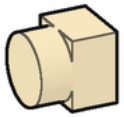
G QT encoder



H 18 / 24 switch



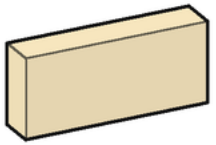
I J2 power switch



J Run button



K Protected pack + XT30



L Princeton USB SSD

Also identify microSD, camera FPC, HDMI heads + ribbon, EVF feed, QT / switch / run / J2 leads, fuse and insulation.

02 Sort hardware and tools

Counts are installed quantities. Purchased spares are not extra installed hardware.

A

7 x PT 3.0 x 12 PH1

Panel 4 | carrier 2 | independent grip lock 1

B

3 x M3 x 10 + washers

Collar anchors into 3 short M3 inserts

C

1 x M3 x 16 + washer

Collar pinch into 1 long M3 insert

D

4 x M2.5 x 5 retained

Upper kit screws; retain 4 original standoffs

E

4 x M2.5 x 8 replacements

Lower screws through carrier; bag 4 old M2.5 x 5

F

2 x supplied cooler push pins

Included in cooler kit; proxy feet are not fasteners

G

1 x tripod nut

Capture in grip before joining the tub

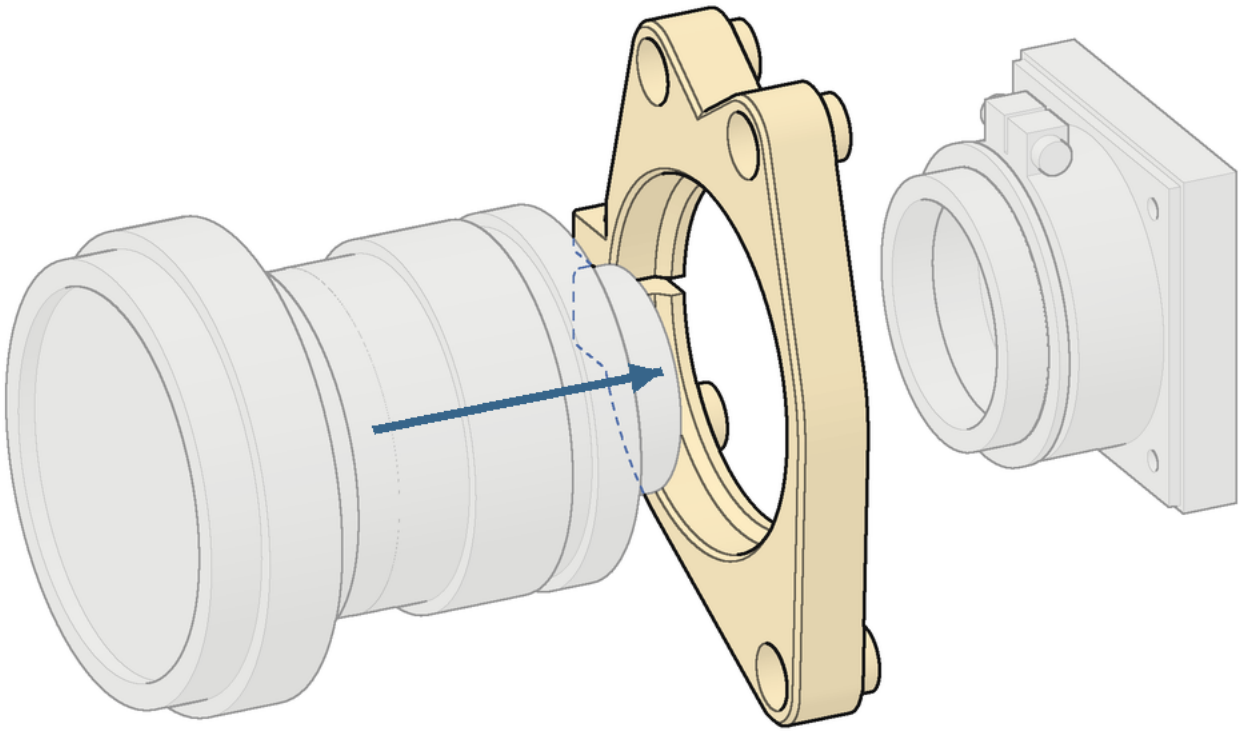
TOOLS

T1 PH1 + torque driver
T2 Iron + correct insert tip
T3 Solder / flux / wick + heat gun
T4 Stripper / cutter
T5 Meter, thermocouple, calipers
T6 Fine tweezers + smooth pliers
T7 Deep sockets, saw, file, soft vise
T8 ESD mat + wrist strap
T9 Panel prop, shims, rule, gauge
T10 Bench supply, Pi PSU, charger
T11 Scope, USB meter, DC load
T12 Insert-repair pin vise / drills

STOP Torque and insert process need revised coupons. No threadlocker on ASA. Power-nut deep-socket fit is unmeasured.

03 Prepare optics and prints

Bench preparation before electronics go inside



Support the load

Support camera and lens separately. Never react lens torque through PCB or cover.

Measure back focus and support band. Keep adapter / BFAR witness marks.

Bag tripod block after setup. Install inserts before electronics.

Keep the camera socket accessible. Seat the Pi cable end before the cooler (04); mate the camera end just before insertion (08b).

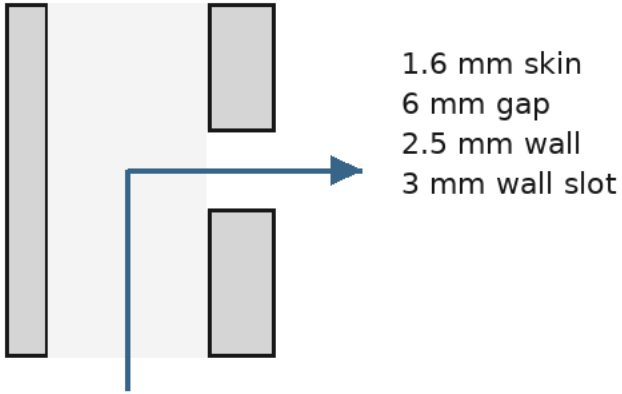
STOP Review the outer plenum on 03v / 03w plus collar and horizontal-bores supports. Measured optics and M1 remain open.

03v Inspect the concealed outer plenum

Representative sections. Blue arrows show intended paths. Windowed ribs and mouth lintels omitted; use the saved print geometry.

COLD BAY / INTAKE

EXTERIOR CAVITY



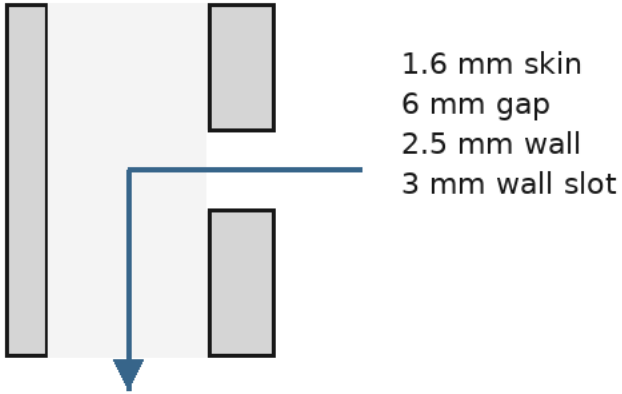
Underside mouth opens into the gap

Rear + underside perimeter mouths feed the cold bay. Four internal wall slots have 420 mm2 nominal total area.

Net mouth necks: rear 333 mm2; underside 370 mm2.

HOT BAY / EXHAUST

EXTERIOR CAVITY



Underside mouth opens into the gap

Four internal wall slots feed the hot bay: 510 mm2 nominal total area. Air is intended to leave at front + underside perimeter mouths.

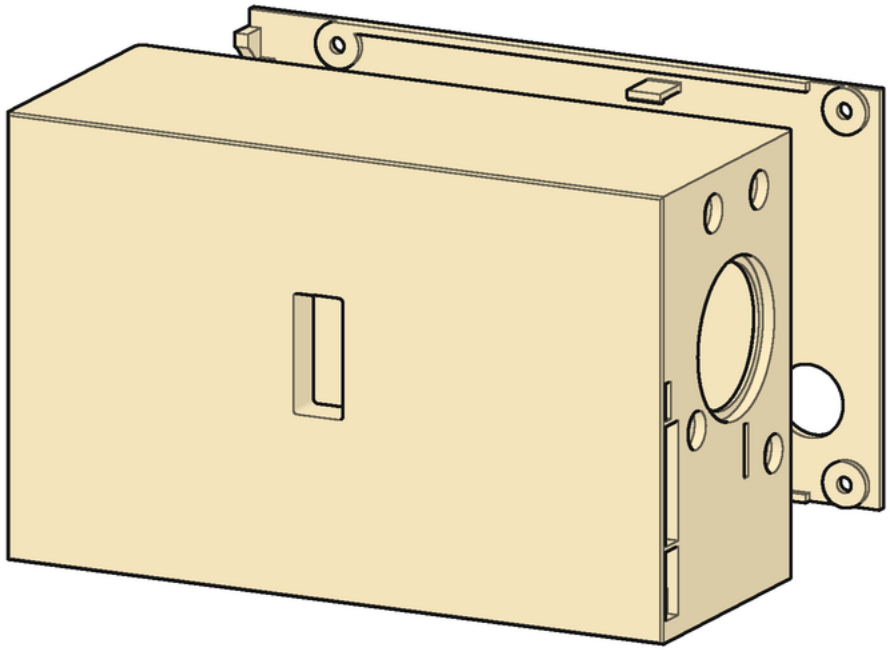
Net mouth necks: front 274 mm2; underside 317 mm2.

Side skin 1.6 mm + gap 6 mm + original wall 2.5 mm. Cavity datum stays Y -25.5. Broad-side grille is concealed; perimeter mouths remain.

STOP CSI is an explicit service aperture. Direct rear/front views expose mouths. No sealed-flow, dust or water rating.

03w Print and inspect the outer plenum

Do not fit electronics or inserts until the full air path can be inspected and cleaned.



PRINT AND INSPECT

1 Tub body STL is -Y-face down. Stub is a separate print. Review each actual print pose and first roof layer in the slicer.

2 Source support: 1.2 mm beam lines on pillars; beam-window spans up to 14 mm. Preserve the declared bridge direction in each roof region.

3 Qualify three coupons: preliminary region, port/service crop and all-four-mouths crop. Geometry-only results do not establish printing or cleanout.

4 No trapped support. Inspect both full-body bays through their intended mouths; remove every loose particle and reject inaccessible debris.

5 Check all four perimeter mouths and eight inner-wall slots. Recheck printed openings, roof sag and actual handling occlusion before electronics.

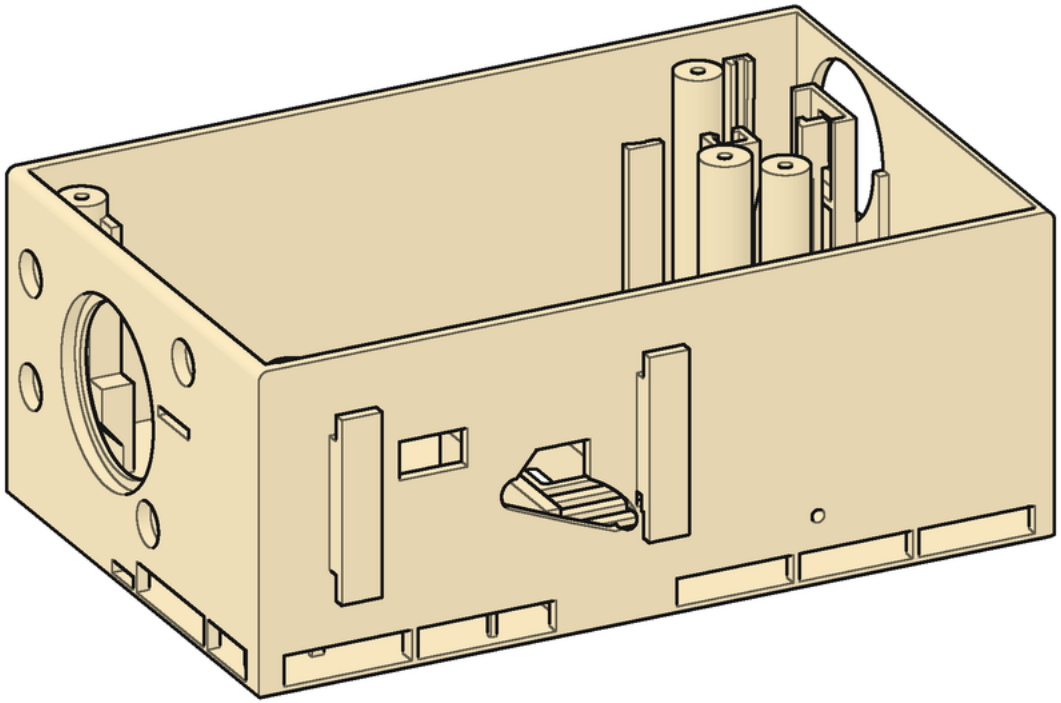
PRINT HOLD: continuous windowed beam lines, pillars, mouth mullions and cold-mouth lintels replace the old post grid. Geometric bridge screens are not slicer or printed-part evidence.

Old panel grooves were blind, with 0.2 mm floors. The revised source specifies a plain 2.8 mm skin.

STOP Blocked slots, roof sag, damaged beams or trapped support are rejects. This plenum has no slicer, bridge, cleanout or physical print pass.

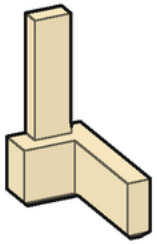
03x Count and qualify the bonded catch

The tub CAD row combines two prints. Bond strength, creep, exact adhesive and actual printed fit remain unqualified.



Source tenon: {'x': [-12.9, -9.8], 't': 1.6, 'h': 10.0}; ceiling floor 1.60 mm. Nominal geometry only.

Piece 1 Tub body in its exported print pose



Piece 2 Catch stub

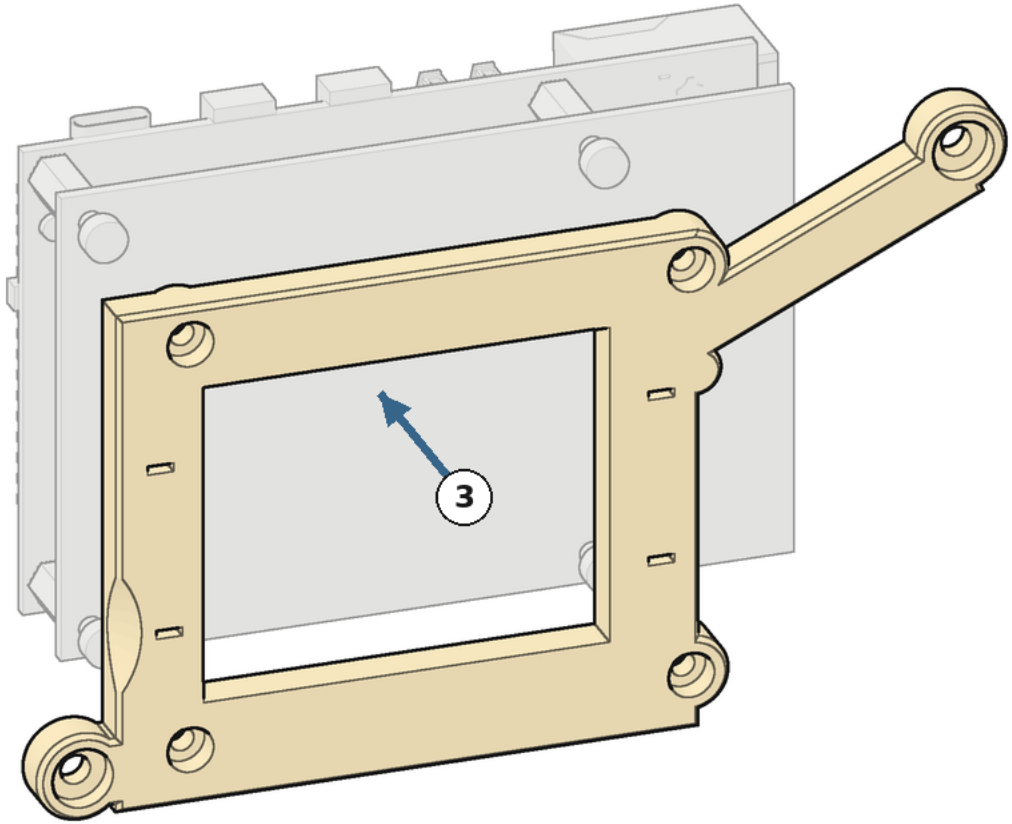
Separate STL and layer direction. Tenon seats against the -Y datum. Keep adhesive off the catch face.

Use the nominated joint dimensions and full stub/pocket coupons. Compare production layer direction, gap/tilt, cure and hot dwell. A beam estimate does not qualify an adhesive or interlayer joint.

STOP No epoxy SKU, print process, loaded service temperature or joint strength is certified. Optical torque and alignment gates remain open.

04 Build one removable stack

Prepare J2, battery-pad and header wiring while both board faces remain accessible.



Keep kit spacing

Seat Pi CSI end BEFORE the cooler. Correct contact face; straight cable and latch. Keep camera end protected.

Then fit cooler with its 2 push pins and native fan lead once.

Retain upper 4 x M2.5 x 5 + original standoffs.

Lower 4 x M2.5 x 8 pass through carrier. Check thread depth before tightening.

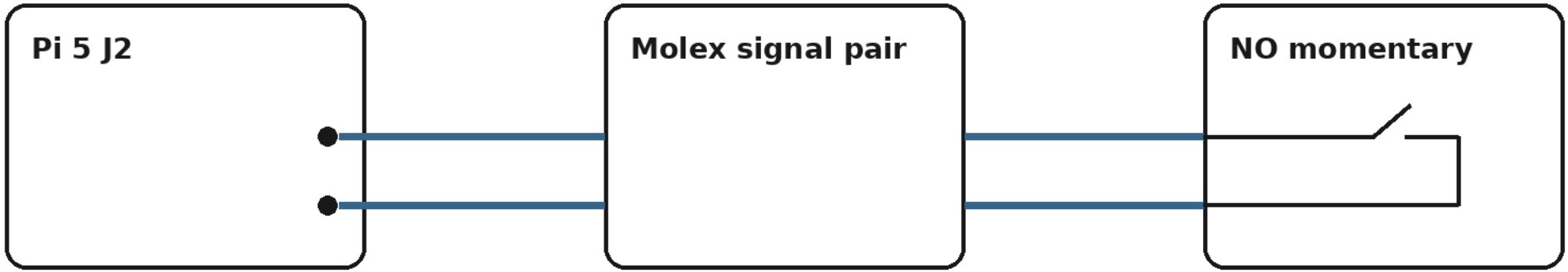
3.4 mm engagement is nominal only.

SOURCE / PROXY HOLD: registered Pi intersects the illustrative cooler perimeter / pin model (79.90 mm3). Measure and register the actual cooler outline before assembly.

STOP No battery. No assumed pad location. Do not change pogo geometry or repeatedly unclip the cooler.

05 Wire the detachable power button

Electrical schematic, not a measured pad-location drawing



Bridge only the two J2 pads with a normally-open momentary switch.

Identify J2 on the actual Pi: between RTC battery connector and board edge. Insulate and strain-relieve all four solder joints. Meter open / pressed continuity before power.

J2 class: Micro-Fit 3.0, 2 circuits, pre-crimped leads. Exact SKU, roots, latch travel and stow remain unqualified. It must not mate to the FPS PicoBlade or EVF RCY halves.

STOP J2 is signal-only, not GPIO or battery isolation. Finished cuts, pad reach, wire exits and nut access remain gates.

06 Prepare control wiring

Physical J8 pin numbers below. This is a pin map, not a rotated-board orientation drawing.

QT encoder

Pin 1 3V3 -> SH 2
Pin 3 SDA -> SH 3
Pin 5 SCL -> SH 4
Pin 6 GND -> SH 1
Bridge A0: address 0x37

18 / 24

Pin 33 GPIO13
Pin 34 GND
CK1044 common A / throw 2
Closed / low = 24 fps
Open / high = 18 fps

Run button

Pin 37 GPIO26
Pin 39 GND

Normally open
Pressed / low = run

Match physical pin 1 to the real board silkscreen / official pinout before plugging.

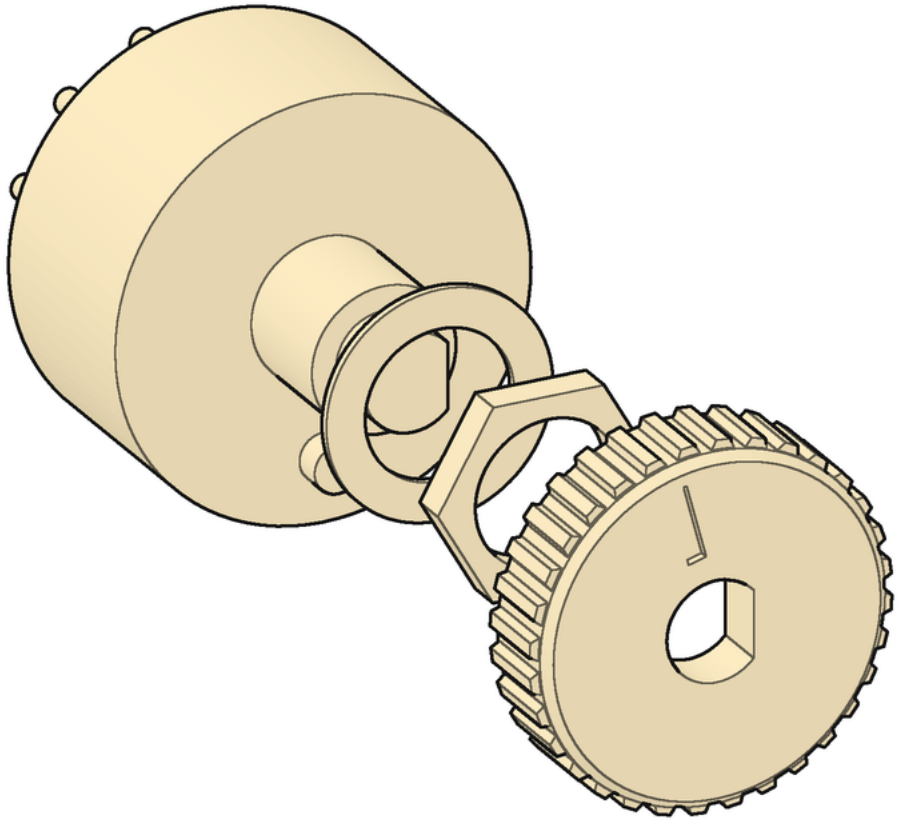
QT target: 250 +/- 10 mm, two source leads, four staggered insulated splices. Meter by SH pin number; never match wire colours alone. Place splice sleeves on a straight, accessible run.

The panel carries encoder and J2. Support the FPS switch on its tub rest; keep its complete pair mated during carrier service. Separate that pair only on the bench.

STOP Header plugs and every wire tail must clear the stack, HDMI, fan and carrier motion.

06b Set the real frame-rate switch

Conditional Lorlin CK1044. Two adjacent detents are 30 degrees apart.



Three distinct pieces

Nut. Mounting washer. Stop / tang washer.

Stop washer shape is not measured and is not drawn. Confirm supplied hardware.

Set two adjacent positions. Meter A / 2: open = 18; closed = 24.

Preserve the stop setting when removing knob, nut and panel.

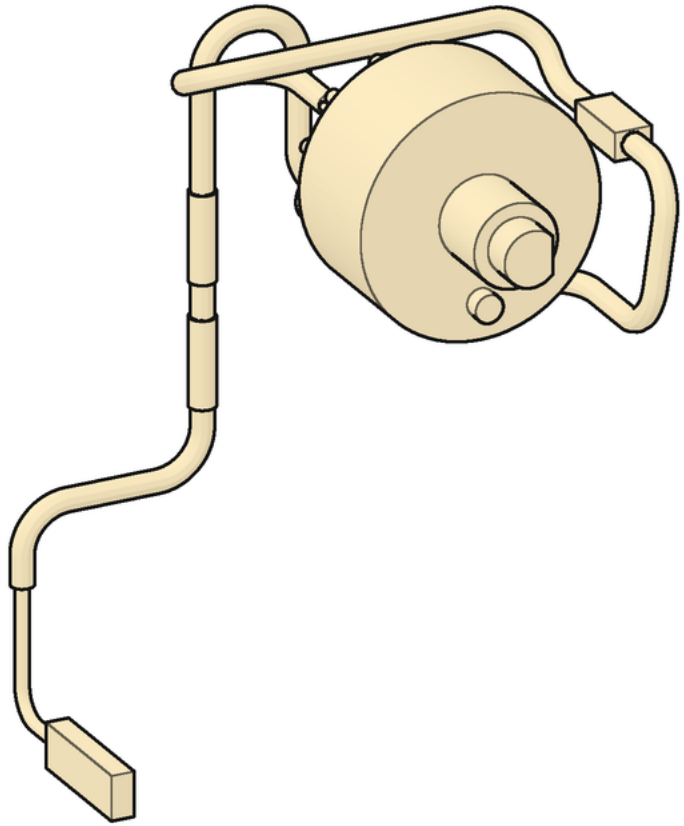
The switch rides with the module; washer, nut and knob go on after the panel (page 16).

Measure the actual stack before cutting. Proposed shaft length 11.8 mm from mounting face is a fit study; 6.35 mm D shaft / 5.5 mm flat needs a real knob coupon.

STOP Nut / mounting-washer shapes are estimated. Actual stop stack, pin clocking, shaft cut and tool access remain gates.

06c Keep the full FPS harness together

Two donor female GPIO heads + two splices + both factory Molex housings + A / throw 2.



Identify both black wires

FPS-only CAD group clears; other looms remain gated.

Male 2181110202: 225 +/- 6 mm exposed stock.

Female 2181120201: 150 +/- 6 mm exposed stock.

200 mm main axis includes donor stubs / splices.

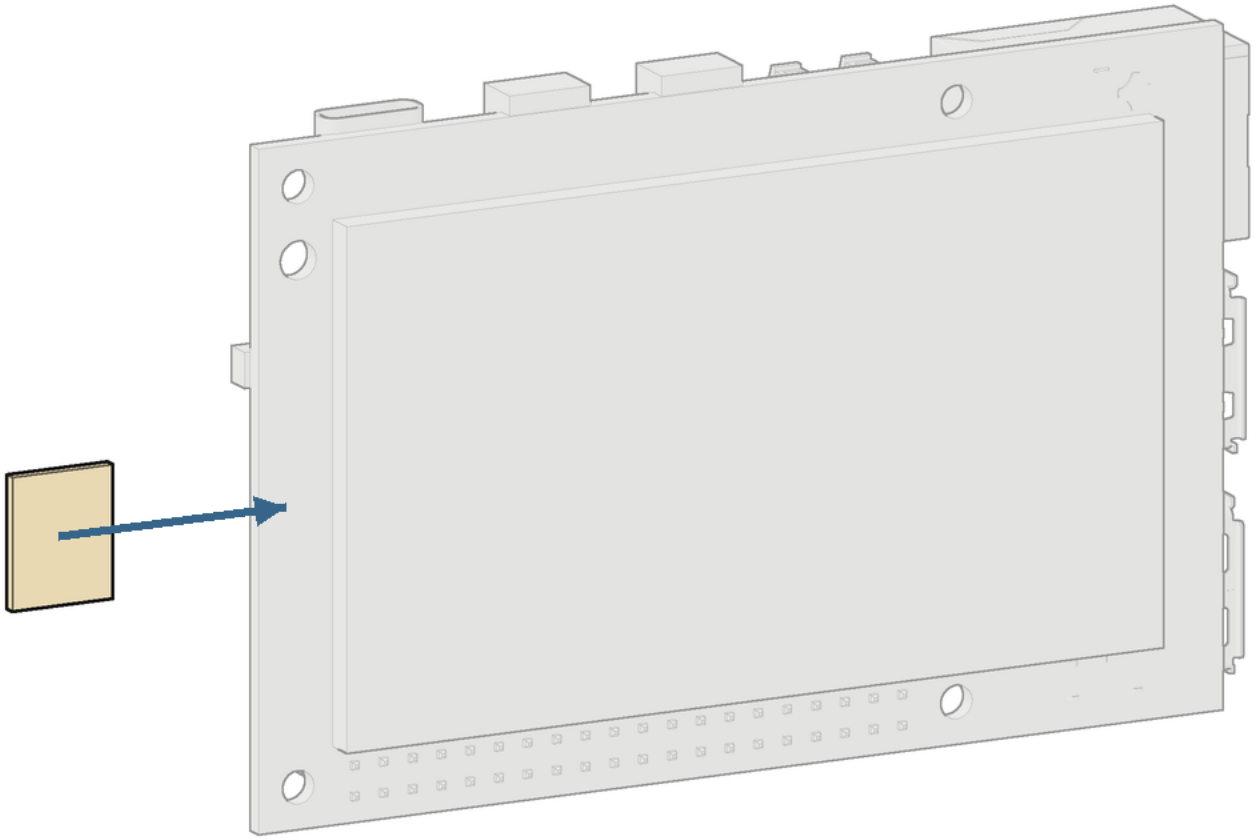
Keep pair mated in the camera. Bench separation only.

Amber cylinders reserve two finished splices, maximum diameter 3 x 10 mm. Cut lengths, solder overlaps, individual-core fanout and terminal joints still require measurement.

STOP Label and continuity-test FPS. J2 uses a distinct Micro-Fit class; exact receipt try-mate checks remain open. Axes are not cuts.

06d Fit the OS card on the bench

The microSD travels with the removable Pi stack. Closed-panel card access is blocked by the power switch.



Card before carrier

Prepare and verify the OS image on the bench.

Identify real contact orientation. Insert -X before the stack enters the body.

Leave the card with the stack during carrier insertion.

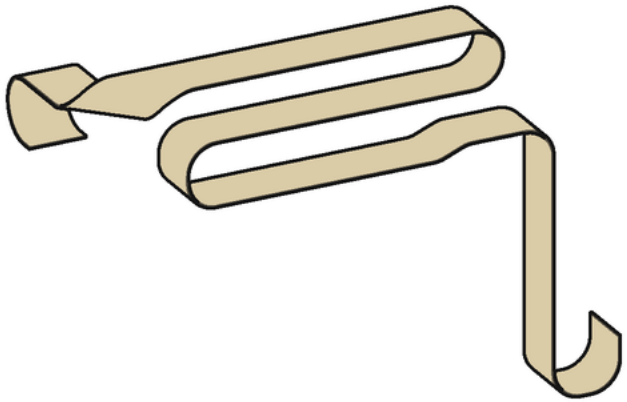
For later card service: isolate all power, remove the panel, then withdraw +X.

The panel-mounted J2 power switch occupies the card extraction route. The front slit alone does not establish service access.

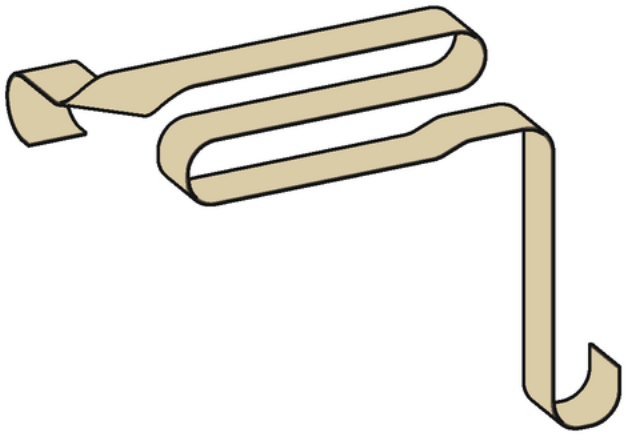
STOP No closed-panel microSD service. Only a panel-off rigid path is checked; real contact orientation and handling remain open.

06e Preform HDMI on the bench

3557 right-angle micro-HDMI at Pi | intact 300 mm 3562 ribbon | 3556 straight head at EVF



Installed



EVF end free / Pi still mated

A full-length study

Each selected pose models 300 mm.

Keep Pi head mated during service.

Both-end-free pose rejected: power-wire conflict.

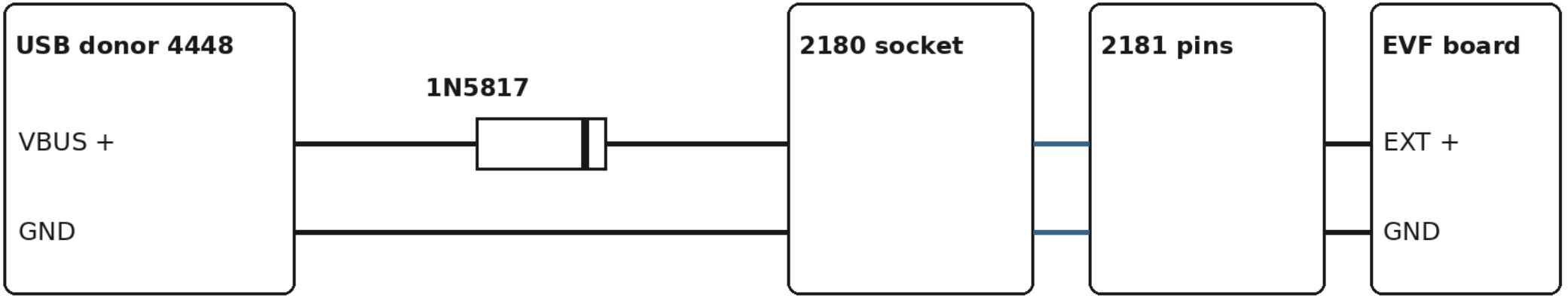
R3 / R5 are allowances, not manufacturer limits.

The complete serpentine strip and both rigid heads must fit through every assembly and service pose.

STOP Full material clears checked solids; head/carry status is separate. Real bend, clocking, retention and EMI remain open.

06f Build the detachable EVF feed

Unpowered electrical schematic. Contact identity is traced by continuity, never colour alone.



Anode: USB side | Band / cathode: EVF side

USB side: 2 diode / positive joints + 1 ground splice. EVF side: 2 traced board joints.

Remove four donor jumper ends only after qualifying the length budget. Separately sleeve D+ and D-; insulate a shield only if present. Keep all joints off bends and strain-relieve both cable ends.

4448 length conflict: catalog 300 mm; supplier drawing 325 +/- 10 mm. Both Generic RCY-style tails: nominal 100 mm / 20 AWG. Final cuts, overlaps and actual plug / junction envelopes remain unmeasured.

STOP Isolate USB and battery. Meter polarity and diode direction. A diode does not guarantee the EVF voltage ceiling; G-W6 remains open.

06g Verify the battery and EVF power paths

Electrical intent only. Bench qualification is required before any full-camera power-up.

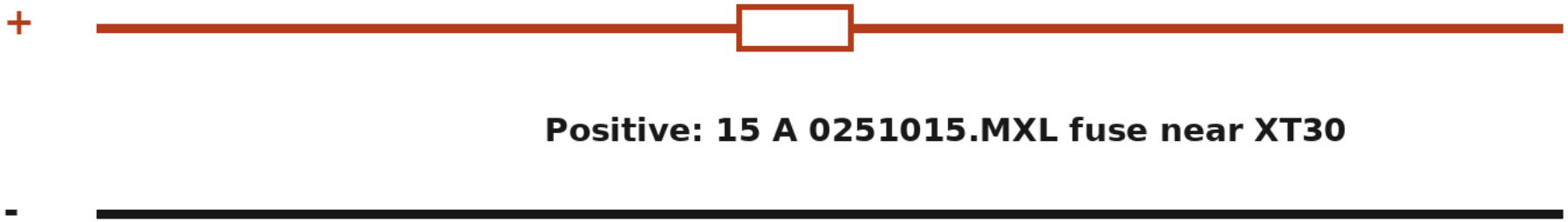
1S2P protected pack
60 mm 18 AWG lead

→

XT30 junction
Mate by housings

→

X1203 battery pads
Pogo feed -> Pi 5



16 AWG: buy 914.4 mm per colour. Both 270 mm path budgets include unresolved pad-root allowances. Positive also includes 20 mm rigid fuse / joint zone. Copper cuts / overlaps TBD.

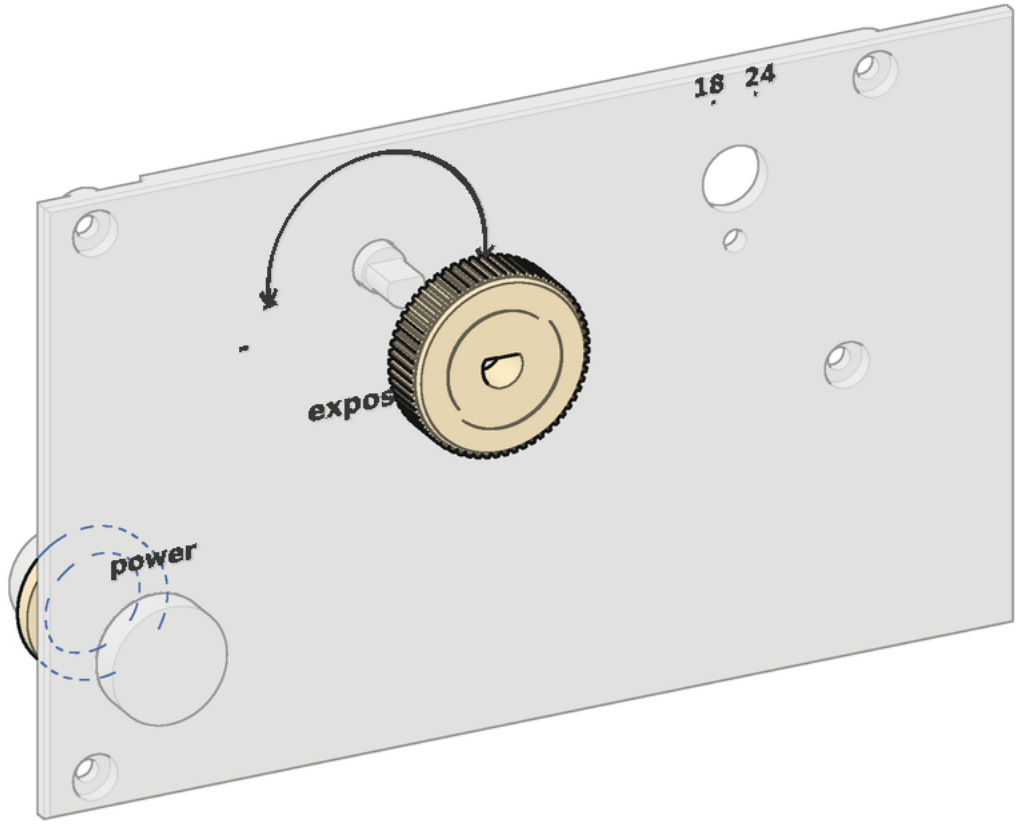
EVF USB positive -> qualified feed -> EXT+; USB ground -> EVF GND.

Inherited diode option: 1N5817 band / cathode toward EVF. It does not guarantee the required voltage ceiling. Regulator alternative is not adopted. Trace the actual EVF revision before soldering.

STOP BLOCKED: 0.14277 to 0.16254 V at 8.6 A versus 0.150 V. Reach and hot-drop need one compatible solution.

06h Build the panel subassembly

Bench. Encoder, exposure knob and power button only; the FPS switch rides with the module.



Thumb behind the board

Snap the encoder into its cradle.

Thumb flat on the back of the encoder board; press knob_exp onto 2 paper strips; pull the paper out.

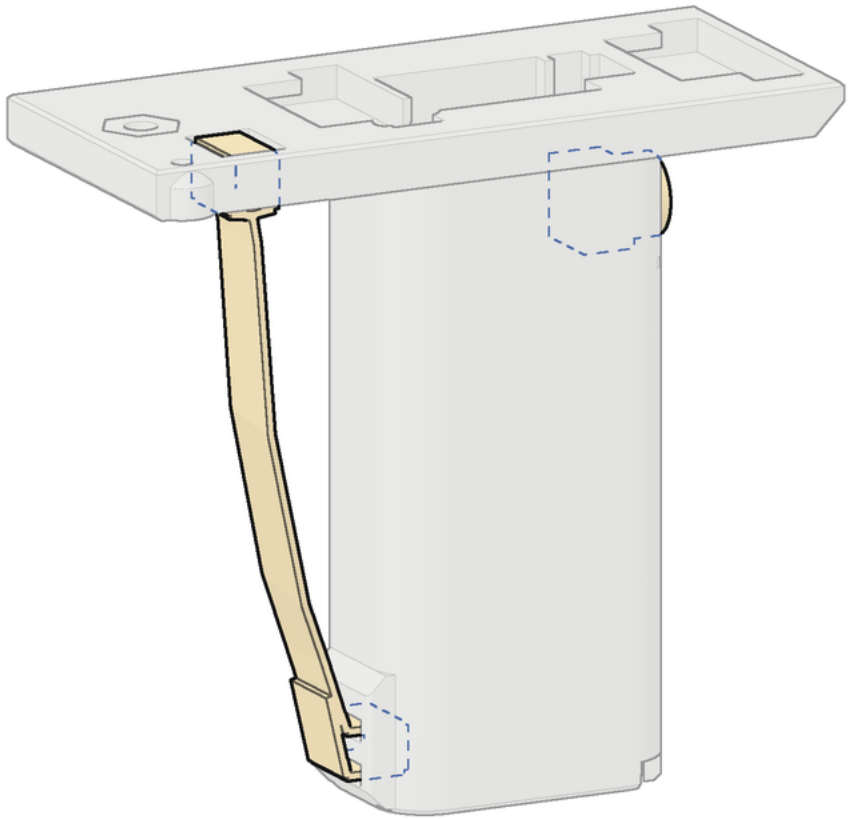
Power button flange on the outer face; nut on the inner face with a deep socket on the button axis.

No FPS washer, nut or knob here.

STOP Thread length and deep-socket reach on the real panel are unmeasured. Keep the J2 junction half free.

07 Prepare the grip, then thread the strap

Battery stays out. Tub off: strap and run lead go in while the base top and floor passage are open.



Strap on the open base

Fit the tripod nut. Feed the run lead up through the open floor passage, then drop the run button into its cradle.

Thread the 12 mm strap through both upper and heel slots now: its pass over the bridge lies in a 2 mm recess in the base top that the seated tub floor closes.

The lower return passes through both heel slots and inside the grip wall. Keep the cap, battery travel and tripod seat clear.

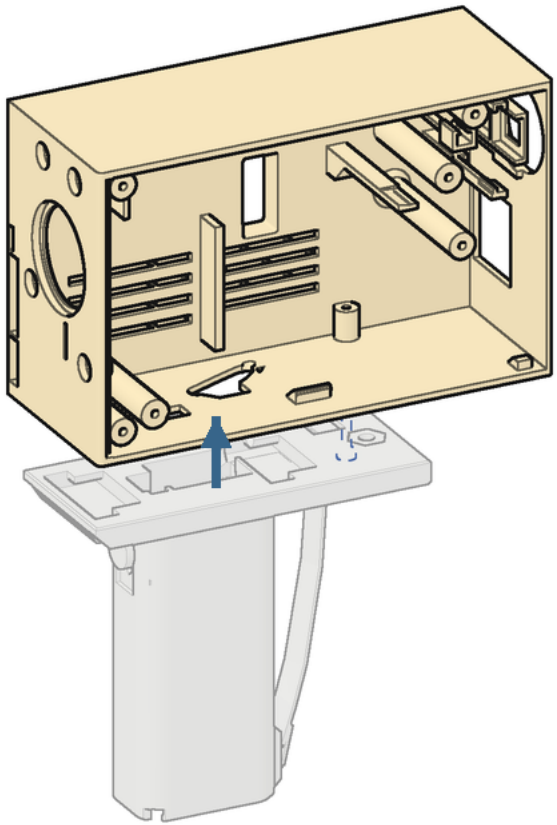
Buckle, closure overlap and retained tail still need real-part layout.

Strap routing proxy only; actual bends, threading, hand clearance and the assumed 1.4 mm thickness need measurement.

STOP Neither anchor nor this routing is a rated safety tether. Qualify the actual strap, closure and both anchors before use.

07b Seat the tub, then lock the grip

Over the threaded strap. Keep the run lead in the floor passage.



Tub on, then lock

Engage the keyhole tongues in their actual keyed direction over the threaded strap; do not lock yet.

The strap is already fitted: it went in on the open base (page 07), before the tub and before the grip-lock screw, whose driver handle passes close to it.

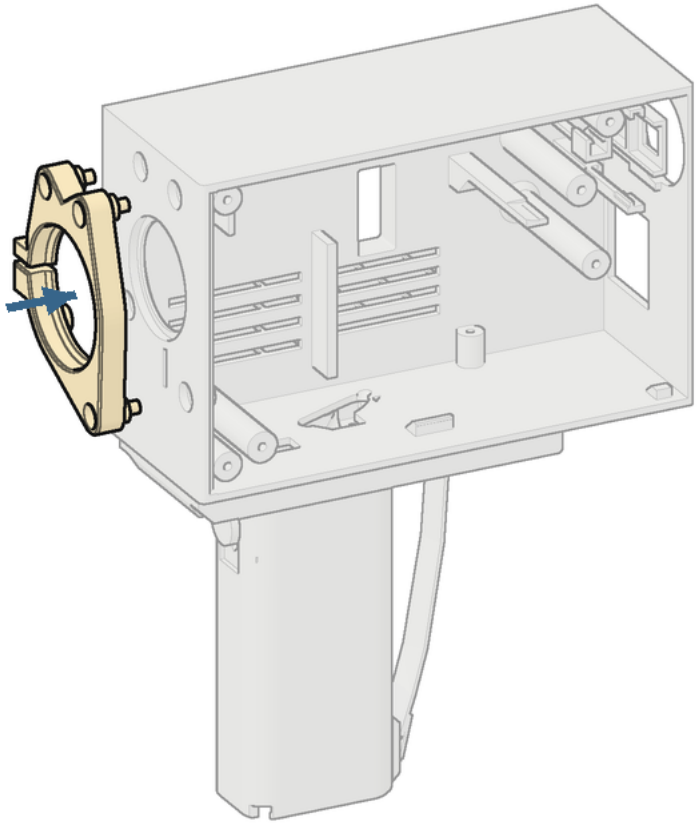
Then drive the 1 x PT grip-lock screw on its axis.

The panel is never the grip lock.

STOP Arrows show approach only. Keyhole engagement, driver access and actual lead passage need dry fitting.

08 Collar and centring gauge first

The camera is still out: the gauge seats in the empty tub lip. Carrier and panel absent.



Gauge, then camera

1 Collar feet onto the tub face over the 3 short inserts.

2 Start s_c1..s_c3 about 2 turns.

3 Push the gauge for this lens through the collar until its rear cone seats in the empty lip.

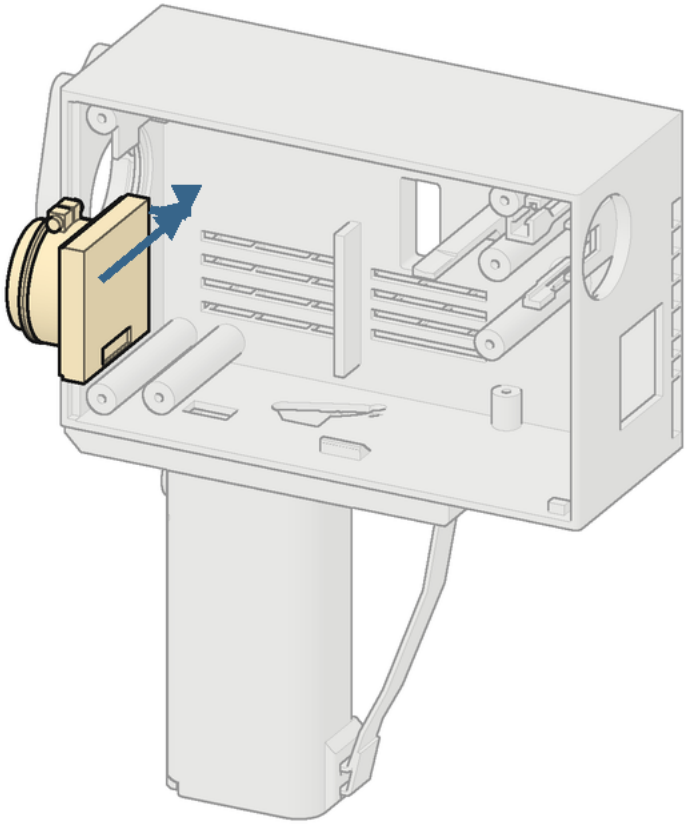
4 Thumb on the gauge; tighten s_c1..s_c3 on axis, 0.15 N m provisional.

5 Pull the gauge out forward.

STOP The gauge and the camera / adapter are never in the body together (CAD overlap 927.75 mm3). The gauge is a tool and is not drawn.

08b Camera before the carrier

The board cannot be pushed through the installed vertical stack. Collar fitted, gauge out.



Three-part motion

Support the prepared Pi module outside. Mate the camera FPC end before insertion; Pi end is already under the fitted cooler.

1 Approach from +Y with rearward / upward offset.

2 Lower onto seating height.

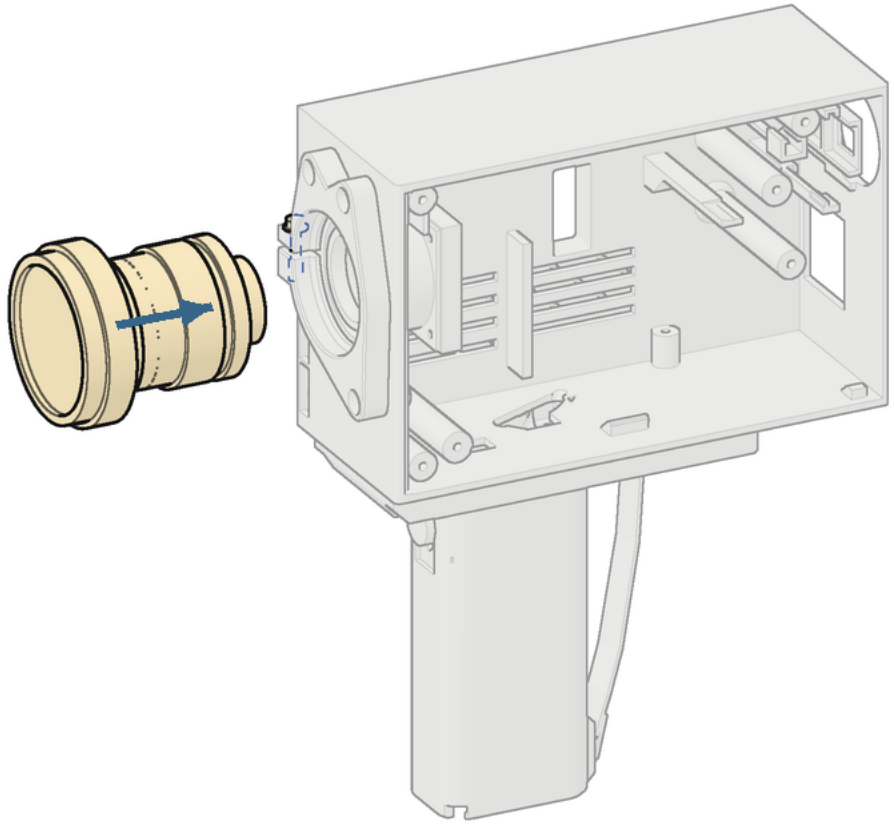
3 Move lensward to datum; lock tab between the tines.

Nothing holds the camera back until the lens is on: go straight to page 08c.

STOP HOLD: rigid camera drawing omits the full 200 mm linked FPC. Finger access, optical support and torque catch remain open.

08c Lens before the carrier and panel

r7 step 8 method through the open +Y side. Fingertips only; never react thread torque through the PCB or cover.



Float, then pinch

Fit s_c4 loosely.

Body on its side, open side up; one fingertip pushes the camera cover forward onto its lip catch.

Screw the lens in with fingertips until it stops; the tab catch takes the torque.

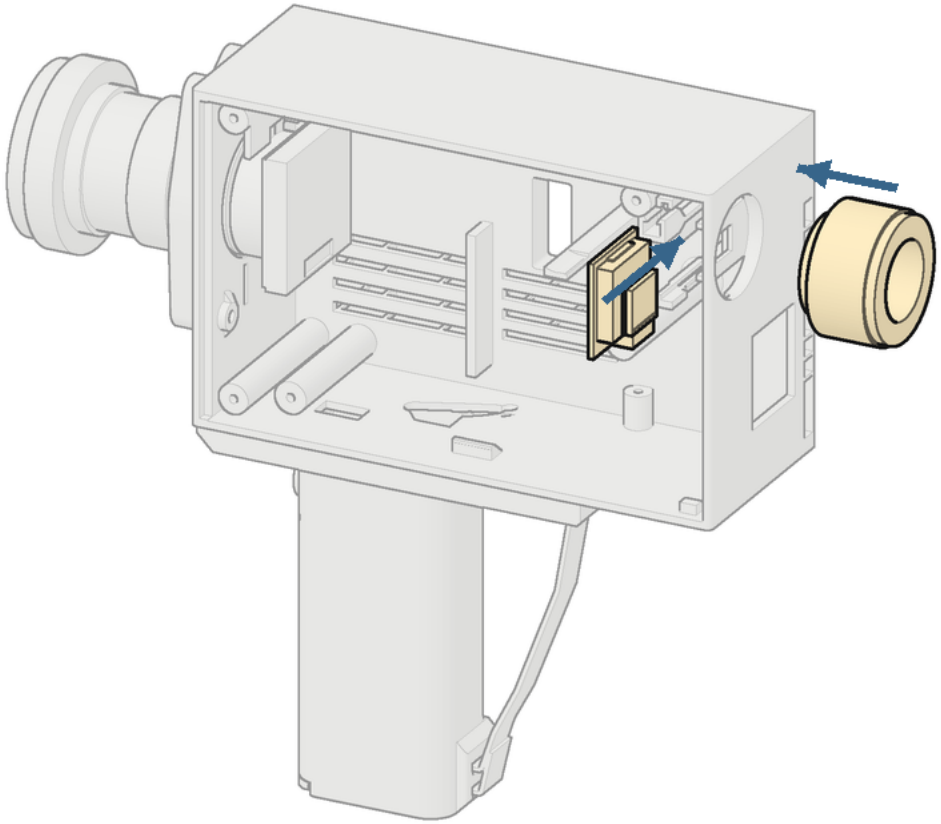
Push the lens onto the collar cone.

With s_c4 still loose, turn lens and camera together the other way until the lock tab stops on the other tine, then back about half way; look and feel: the camera touches nothing. Then snug s_c4 on its axis, 0.2 N m provisional.

STOP Source-bound roll / J7 geometry does not qualify printed float, bond strength or optical alignment. Check both witness marks.

09 Eyepiece, then the EVF module

Keep OLED foam, native flex and connector latches unstressed.



Keep the board secure

Push the eyepiece spigot +X through the rear-wall bore from behind first.

Slide the module from +Y into its seats; foam and native flex unstressed.

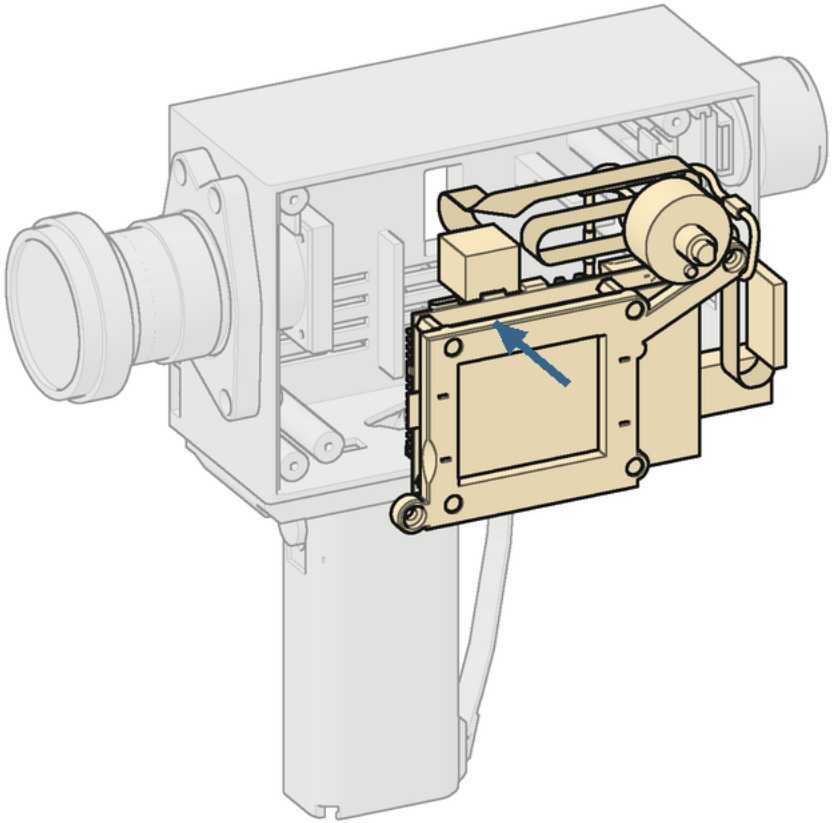
The panel supplies clamping later; do not let the open module fall.

No HDMI or EVF power mating yet: their ends arrive with the carrier.

STOP Native flex contact face follows the actual EVF kit. Do not infer it from a CAD envelope.

11 Insert the stack with a free EVF end

Intended trial sequence: Pi HDMI adapter and OS microSD fitted first; recording drive absent.



Hold before proceeding

Run lead onto Pi pins 37 / 39 at the staging pose.

Carry FPS bushing and stack together from +Y, both supported.

EVF HDMI end free, at least stroke + 3 mm low.

Pigtail through the floor opening; FPC clear of the 3.9 mm gap.

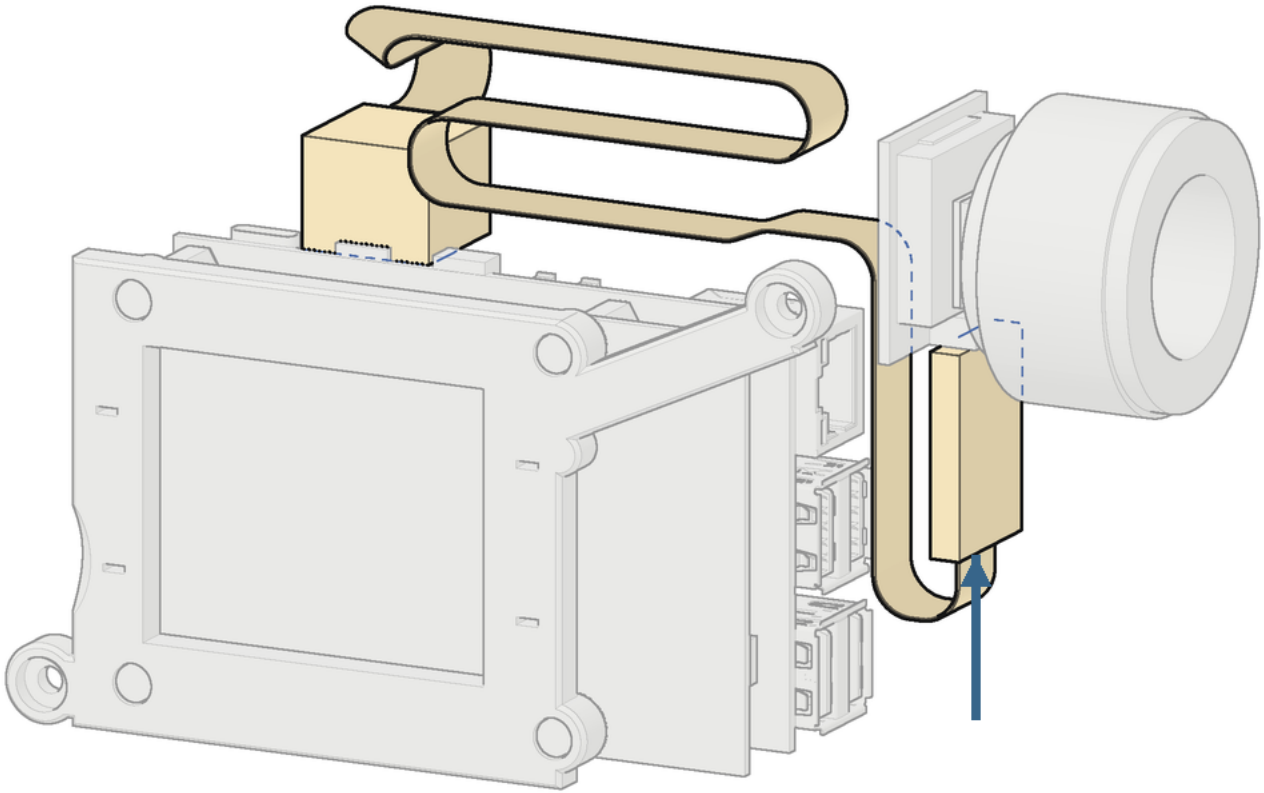
Seat carrier; 2 x PT screws.

CAD: full HDMI and carried heads checked at 2 mm poses, with 9 mm head drop (6 mm stroke + 3 mm margin). Other looms remain gated.

STOP HOLD: complete 200 mm CSI ribbon is already linked at both ends and is absent from the rigid carrier sweep. Full flexible motion is unproved.

12 Mate the EVF ends from inside

Step 5, fully unpowered. Storage stays out. Hold the adapter body, never pull the ribbon.



EVF head upward

Bring the straight 3556 head from below.

Confirm the keyed receptacle and actual contact face.

Mate fully by its body. Add qualified strain relief clear of the latch.

No twist to correct a wrong head.

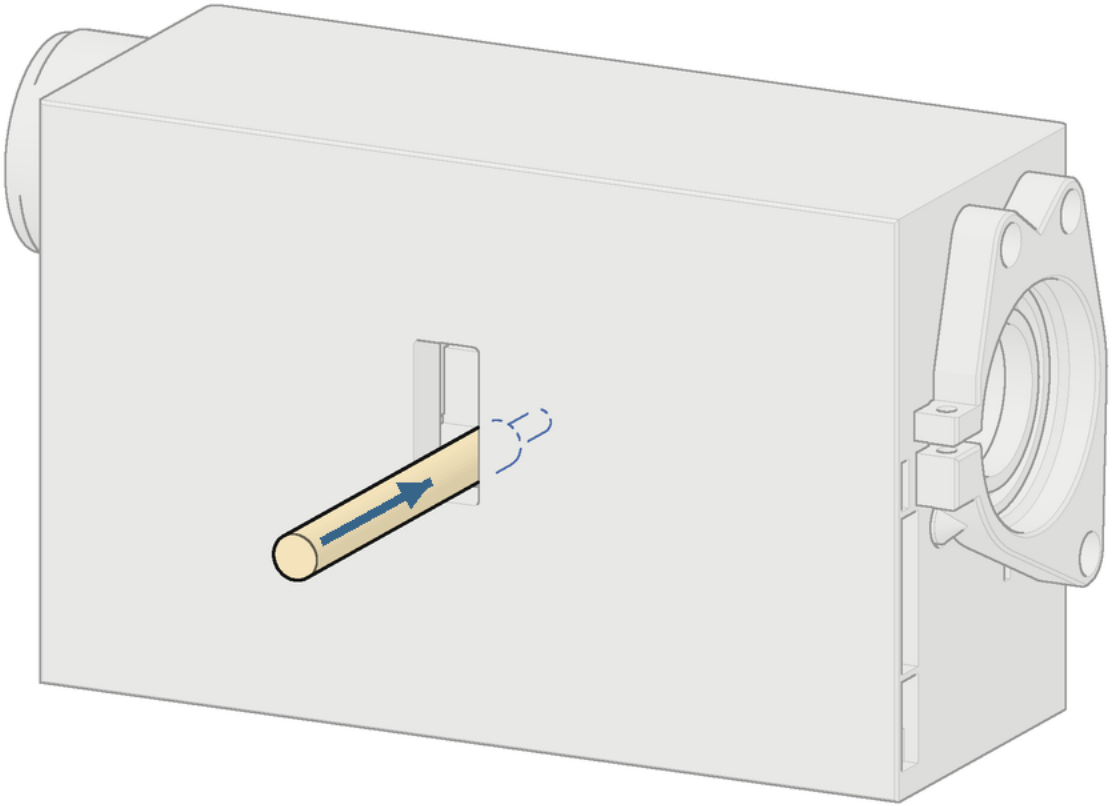
Then mate the EVF power RCY pair by both housings.

Tub omitted for visibility. Adapter blocks are unmeasured catalog envelopes, not detailed keyed connectors.

STOP Upward head travel, fingers / pliers and simultaneous USB + HDMI access must all clear. Final pose is insufficient.

13 Prove camera connector access

Pi end seats before cooler; camera end before camera insertion. This is a tool-reserve view, not an instruction to mate late.



Keep the complete ribbon

Pi end BEFORE cooler (04). Camera end supported before insertion (08b).

Both ends are linked when the carrier moves. The rigid sweep omits all 200 mm of FPC.

Prove full width, roots, slack, actual latch/jaws and motion with other looms.

Side cylinders do not qualify service with the cooler retained.

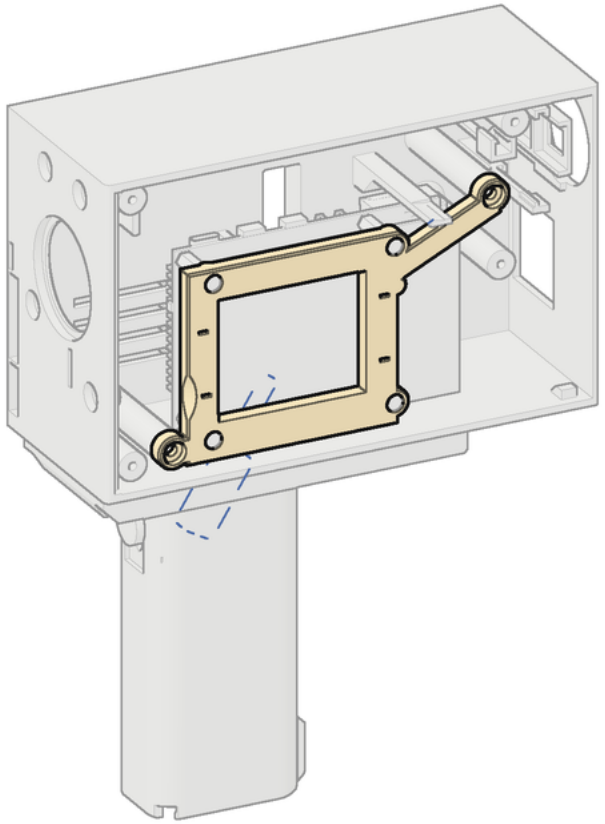
Rigid cylindrical reserve clears only; actual tool, cooler and full FPC remain unresolved.

Intake wall slots: 420 mm2 nominal. CSI remains a straight unsealed bypass; keep latch/tool access clear.

STOP OPERATIONAL HOLD: actual cooler outline, latch operation, inserted FPC and interacting flexible looms are unresolved.

15 Prove stored slack and disconnection

Complete cable lengths, connector bodies, sleeves and tie heads must fit at the same time.



No hidden excess

Static wire capsules: clear printed solids; root joints omitted.

Keep fuse straight and supported. Keep joints off bends.

Four tie slots are provisions, not a tested retention scheme.

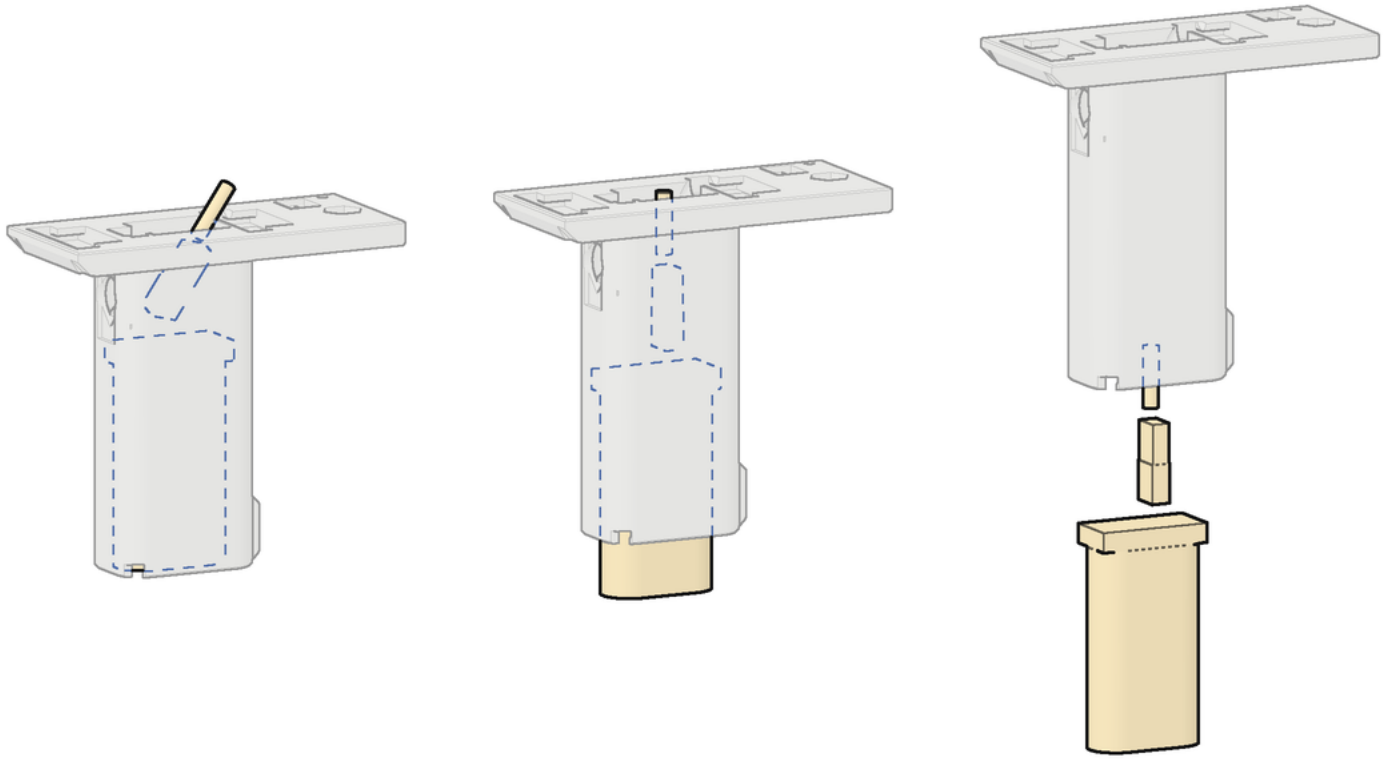
Loops must clear CSI tools, GPIO tails and UPS PCB.

Test pack extraction with retention unchanged.

STOP If XT30 cannot be parted before opening the body, the assembly is not safely serviceable. Redesign before closure.

15b Rigid connector withdrawal study

Body omitted for clarity; it stays closed during isolation. Full flexible-harness service is still blocked.



Stored **Unwind / lower** **Withdrawal limit**

What changed

CAD checked: 201 sampled poses clear the closed assembly, with pack moving and cap removed.

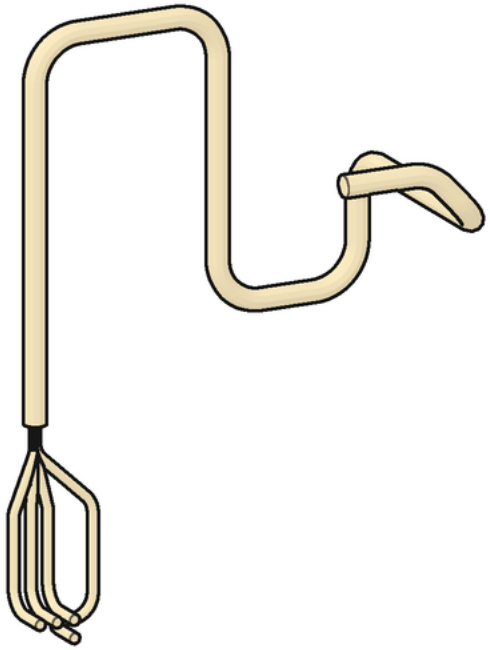
The pack moves with the junction.

This does not prove the complete conductor paths, joints and retained slack can deploy or return safely.

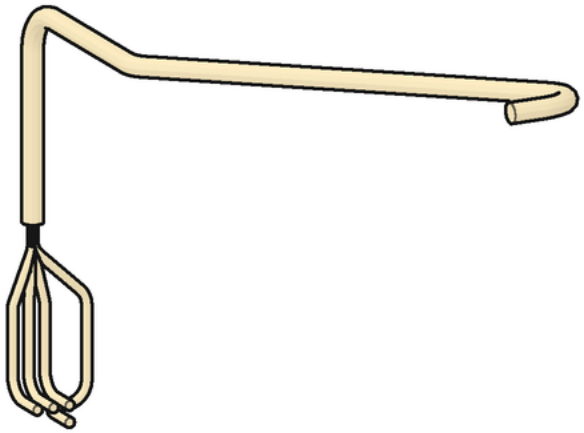
STOP Do not pull the pack through a blocked loom. Complete wires, pad joints, retention and pinch clearance remain STOPs.

15q Separate QT storage from free mating reach

Source-bound routing study. The allowance is a reach budget; full flexible transition and real cable packing remain HOLD.



Stored nominal 250 mm lead



Free curve at propped panel

Stored: main 206.93 + longest branch 43.07 = 250 mm nominal. The 240-260 mm family is checked separately.

Free reach: curve 147.47 + branch 43.07 + 24 mm allowance. Remaining stock is not drawn as a qualified flexible shape.

Interpret the scope

Round routing envelopes only.

Four 1.6 mm cores cannot pack inside the 3.2 mm bundle proxy. Real MP-QT section, sleeves and roots are unqualified.

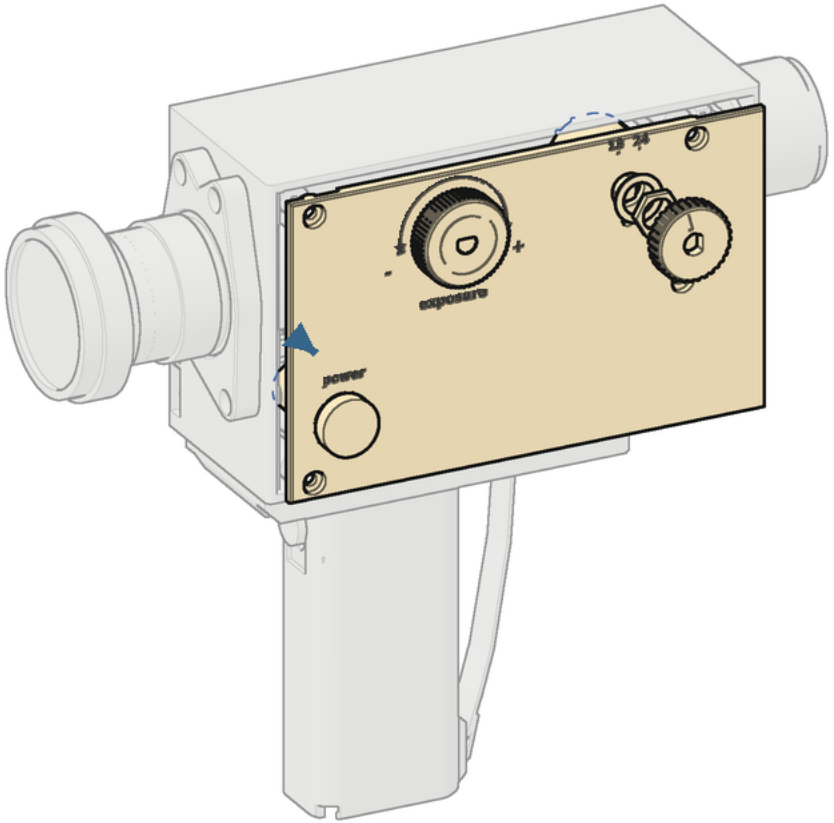
A minimum-reach pass cannot prove all cable stows or moves between these poses.

Keep four splices and finished 250 +/- 10 mm intent.

STOP No cut instruction or flexible-motion pass. Keep purchased parts and finished-length range unchanged.

16 Close the panel, then fit the FPS hardware

Step 6. Shown part-open; service removal is the exact reverse. Keep the FPS pair mated.



Panel, screws, then FPS

Prop the panel; mate QT and J2.

Move it in along -Y, guiding bushing and peg; support the switch body.

Drive the 4 panel screws before the FPS knob is on.

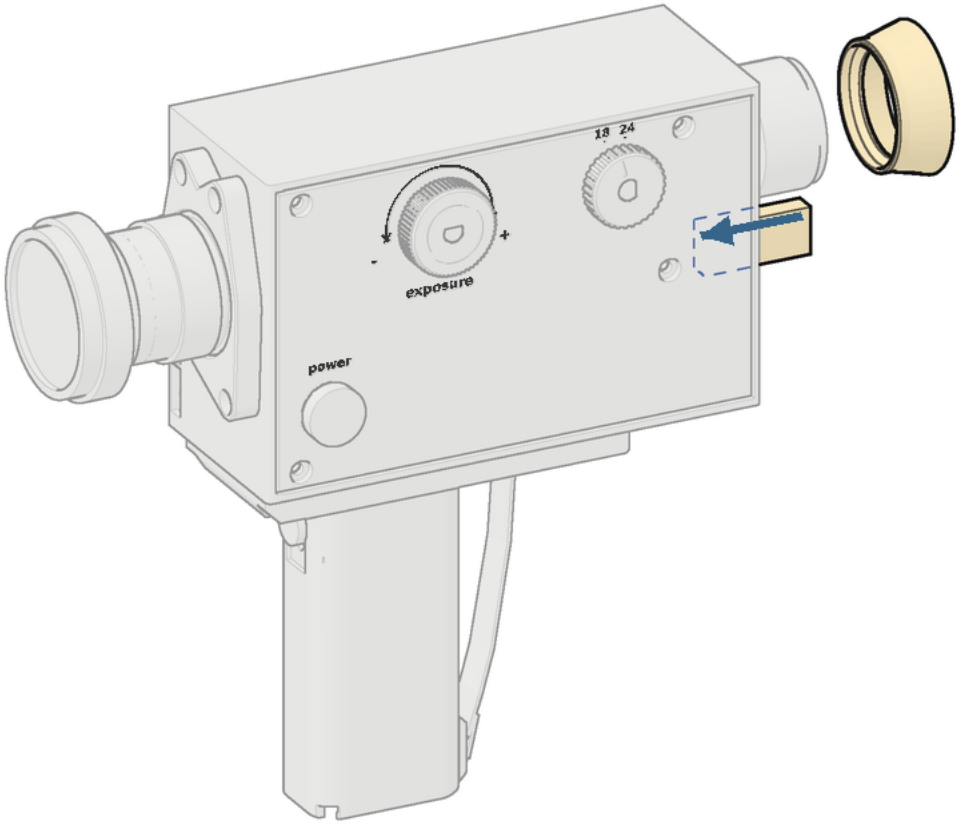
Then washer, nut (deep socket on the bushing axis), knob on the paper strips.

Removal: knob, nut, washer, screws; tip support for the first 8.5 mm, then bushing.

STOP Never use screws to crush slack into the enclosure. Real shaft grip, stop-stack retention and whole-harness motion remain STOPs.

17 Recording drive and eyecup

Fit selected compact Princeton drive with its protective cap removed. OS microSD is already installed.



Straight insertion

Drive travels +X into upper USB3.

OS microSD was fitted on the bench. Later card service requires panel removal.

Push the eyecup over the eyepiece barrel from the rear.

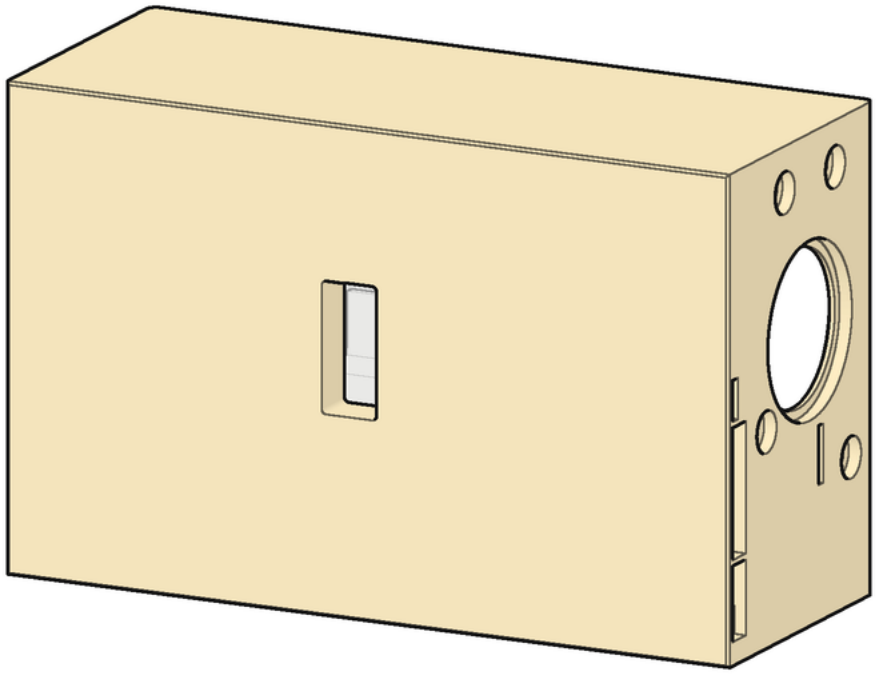
The lens and s_c4 went on in step 2 (page 08c).

Estimated storage-plug keepout omitted: it is not a physical inventory item or a proven mating shape.

STOP Published Princeton size includes cap. Real plug fit and warm sustained >=100 MB/s are not measured.

17v Keep ventilation clear during assembly

The outer bays conceal the inner slots. Their pressure loss, leakage, recirculation and cooling remain unmeasured.



Tub, stack and illustrative cooler. Panel and looms omitted for visibility; prove clearance with the complete final assembly.

Saved-STEP checks: 49 pass / 0 fail / 21 info. Path-aware hot min cut: 510.0 / 510 mm2 (pass). Mouth totals are not a flow result.

BEFORE CLOSURE

Keep mouths, inner slots and baffles clear of full FPC, EVF feed, power loop, tie heads and fan lead.

CSI stays 12 x 27 mm R1 and uncovered. It is an unsealed cooling bypass.

Test hand, strap and stand occlusion. Blocking the hot underside leaves only the front neck: 274.2 mm2 versus 510 mm2 wall slots. No blocked-vent operation is qualified.

THERMAL GATE

After electrical prerequisites pass, record ambient, closed-body load, fan speed/duty, component temperatures and throttling at normal hand positions. Compare with the open reference using the same hardware and conditions.

Measure the actual cooler first; its proxy/source conflict remains open.

STOP No airflow, fan operating-point, hand-occlusion or thermal pass. All 56 inherited physical gates, M1 and candidate gates stay open.

18 Close the loop before power

Physical checks and measured records decide whether this candidate can become a build.

- [ ] Pack disconnected during all internal work
- [ ] Connector identities, contact faces and continuity verified
- [ ] No trapped ribbon, fuse sleeve, splice, tie head or fan lead
- [ ] Panel flush; grip lock, strap and optical support secure
- [ ] Plenum mouths and wall slots clear; no trapped support; CSI open
- [ ] XT30 disconnect reachable with retention fitted
- [ ] Required bench gates passed before pack is connected last

Then qualify closed-body power, thermal, storage and recording performance, plus revised collar / strap loads. Nothing in this booklet closes a physical gate.

STOP 56 inherited gates + M1 + candidate gates remain open. No fabrication, safety or performance certification.

19 Service in dependency order

Always start with page 00 and physical battery disconnection. Routes and refits: ASSEMBLY-VERTICAL section 5.

Panel	Knob, FPS nut (deep socket), washer, 4 screws off -> tip, then bushing support, +Y -> QT / J2 apart (one J2 cycle). Refit: panel, screws, washer, nut, knob.
Pi module	Drive out -X -> panel off -> EVF HDMI / power and Pi FPC released -> 2 carrier screws -> stage +Y, switch supported; run lead apart there; pigtail drawn up.
Lens, same collar	Closed body: s_c4 one turn, nose-down, fingertip unscrew. Refit nose-down; turn back to float; then s_c4.
Lens, other collar	Full chain: lens, panel, drive, module, camera, then collar. Refit from that collar and its gauge (lip empty), camera, lens float, module, panel, drive, pack.
Camera	Lens, panel, drive and the whole module out first (VN-42, user decision); reverse camera path. Refit camera, then lens, before the carrier.
EVF	Panel off -> EVF power pair and HDMI body unplugged -> flex protected -> module +Y.
OS card	Panel off before in-body microSD withdrawal, or on the bench with the module out.
Grip	Pi module out first (run lead at pins 37 / 39) -> grip-lock screw -> grip off.
Refit	Collar + gauge before camera. Lens + s_c4 before carrier. Junctions before panel. FPS hardware after panel screws. Pack last.

STOP CSI tool clearance is not latch qualification. GPIO plugs stay carried; actual FPC release, run slack and wire withdrawal must pass.

19b Camera and collar service: a user decision

VN-42. The current geometry needs the whole module out before the camera can leave.

Option A Accept

Camera, collar and any lens change that needs a different collar go through the full module route: panel off (one J2 cycle), module STOPs, camera out. A collar change always removes the camera because the centring gauge needs the empty lip. r7 needed only the pack out and the panel off.

Option B Redesign

B1: let the camera leave ahead of the carrier front, so camera service needs only the panel, as in r7.

B2: a centring gauge that seats without entering the lip, so a collar can be re-centred with the camera in place.

Either changes the service model and needs new motion checks; neither is modelled.

STOP Not decided in this guide. Pages 08 to 08c and 19 show option A, the current geometry.

20 Evidence and open checks

Source-linked review candidate. Vendor facts are not real-part measurements.

Snapshot

Mesh manifest: 294220bc7e1babc5. No build time is printed, so the same inputs give the same pages.

Geometry: compact-kowa_lm6hc meshes plus source-bound concealed-vent STEP checks; 300 mm HDMI snapshot. Wiring intent: candidate assembly/BOM and WIRING-VERTICAL. The separate receipt records exact source hashes.

What the CAD motions check

Rigid body motions are sampled, not continuous swept-volume proofs. Complete HDMI shapes and carried heads have a separate source-bound check. The source-registered CSI ends have a limited declared-tool check; real cooler, latch and full FPC access remain blocked. Other flexible leads, hands, tolerances and physical retention remain unqualified.

Primary sources and 12 October evidence

Official camera connection order

Pi right-angle head 3557

EVF straight head 3556

Intact 30 cm ribbon 3562

Compact Princeton SSD

Lorlin CK rotary switch drawing

USB donor 4448 + supplier drawing

EVF source socket half 2180

EVF pin half 2181

Molex male assembly drawing

STOP No manufacturer bend limit, purchased cable measurement or production release is invented here.